Capstone Project Specification

AI-Powered Enterprise Helpdesk & Workflow Automation System (AI-HAS)

Target Audience: 4th Year BS IT	**Duration:** 8 Weeks (2 Months)	**Team Size:** 3–4 Members

1. Executive Summary & Project Overview

The AI Helpdesk Assistant (AI-HAS) is an enterprise-grade IT Support Automation platform designed to modernize level-1 operational support. By combining Retrieval-Augmented Generation (RAG) with Function Calling and Webhook Automations, the system resolves routine IT requests (password resets, software access, VPN troubleshooting), routes complex issues to human agents, and logs actionable metrics.

Key Project Objectives:

- **Primary Objective:** Minimize MTTR (Mean Time to Resolution) and reduce L1 ticket load through grounded automated responses.
- **Accuracy & Guardrails:** Prevent hallucinations via dynamic RAG context retrieval and enforce data privacy/safety rules.
- **Workflow Automation:** Execute operational tasks automatically via LLM function calling and administrative API triggers.

2. Real-World Reference Applications & Industry Benchmarks

Students should study existing industry models and open-source implementations to understand operational standards, UI/UX flows, and system integration patterns:

- **Chatwoot + Captain AI (Open-Source Architecture):** Chatwoot is an open-source customer support platform (Ruby on Rails/Vue.js). Its native AI module, Captain, uses RAG for KB answers and automated triage. Serve as an architectural reference for open-source database schemas, webhooks, and live chat widgets.

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- **Zendesk AI Service Desk (Enterprise Leader):** Industry-standard platform featuring intelligent ticket triage, automated agent copilot, and autonomous resolution workflows for software provisioning and access requests.
- **IBM Service Desk with Agentic AI (Enterprise Automation):** Focuses on intent recognition and multi-step agentic workflows that trigger administrative API calls (e.g., Active Directory password resets, Jira incident creation).

Reference System	Architectural Focus	Capstone Relevance
Chatwoot (Captain)	Open-source, webhooks, live-chat UI	Backend schema, webhook triggers & Frontend UI design
Zendesk AI	Knowledge management & action workflows	RAG pipeline & Function Calling integration logic
IBM Service Desk	Agentic AI, safety guardrails & triage	Testing, Guardrails & LLM Observability dashboard

3. Technical Architecture & Stack

Component	Recommended Technologies	Key Responsibilities
LLM & Orchestration	OpenAI API (gpt-4o-mini), LangChain / LlamaIndex	Intent classification, RAG response generation, function calling extraction.
Vector Database	Pinecone, ChromaDB, or Qdrant	Storing embeddings of IT Knowledge Base SOPs and document retrieval.
Backend API	Python (FastAPI) or Node.js (Express)	API endpoints, business logic, RAG orchestration, webhook handlers.
Frontend UI	Next.js, React, or Streamlit	Conversational chat portal, Admin KPI dashboard, KB management UI.

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Automation & Integration	n8n, Make, Webhooks, Jira/ServiceNow APIs	Automated ticket logging, password resets, and human escalation workflows.

4. Recommended Team Roles (3-4 Members)

Role	Core Responsibilities
Member 1: AI / RAG Engineer	Vector database indexing, embedding pipeline, system prompt optimization, guardrails, RAG context retrieval evaluation (Faithfulness, Relevance).
Member 2: Backend & Automation Lead	RESTful API development, database schema management, integration with Jira/ServiceNow endpoints, function calling execution handlers.
Member 3: Frontend & UI/UX Developer	Interactive chatbot widget, agent human-in-the-loop dashboard, response streaming, analytics visualizations (ticket deflection, latency).
Member 4: DevOps, QA & Documentation	CI/CD setup, Docker containerization, performance load testing, synthetic test script suite, final technical documentation lead.

5. 8-Week Implementation Roadmap

Timeline	Milestone Phase	Key Deliverables

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Weeks 1–2	Foundations & Architecture	System architecture design, DB schema, GitHub/Docker setup, and prompt taxonomy prototype.
Weeks 3–4	RAG Pipeline & Core Backend	Vector DB indexing of SOPs, hybrid search retrieval, REST API endpoints with grounded citation.
Weeks 5–6	Automation & Dashboard UI	Function calling integration (password reset, Jira ticket creation), user chat interface, and admin dashboard.
Week 7	Guardrails, QA & Testing	PII redaction filters, prompt injection guardrails, system evaluation audit, and dockerized build.
Week 8	Capstone Defense	Final technical documentation, presentation deck, and live defense demonstration.

6. Grading Rubric

Evaluation Criteria	Weight	Assessment Focus
Core AI & RAG Architecture	**25%**	Accuracy of vector search, dynamic prompt construction, hallucination reduction, and context citation.
Software Engineering & Automation	**20%**	Clean architecture, API design, effective implementation of LLM Function Calling and automated webhooks.
Governance, Security & Guardrails	**15%**	PII scrubbing, prompt injection prevention, RBAC, and system resilience.
Evaluation & Testing	**15%**	Metrics-driven validation (RAG accuracy, latency, ticket deflection rate, system stress testing).

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Technical Documentation	**15%**	Completeness of architecture diagrams, API specs, user manuals, and setup instructions.
Final Presentation & Live Defense	**10%**	Quality of demonstration, defense against edge-case test queries, and team Q&A response.

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